

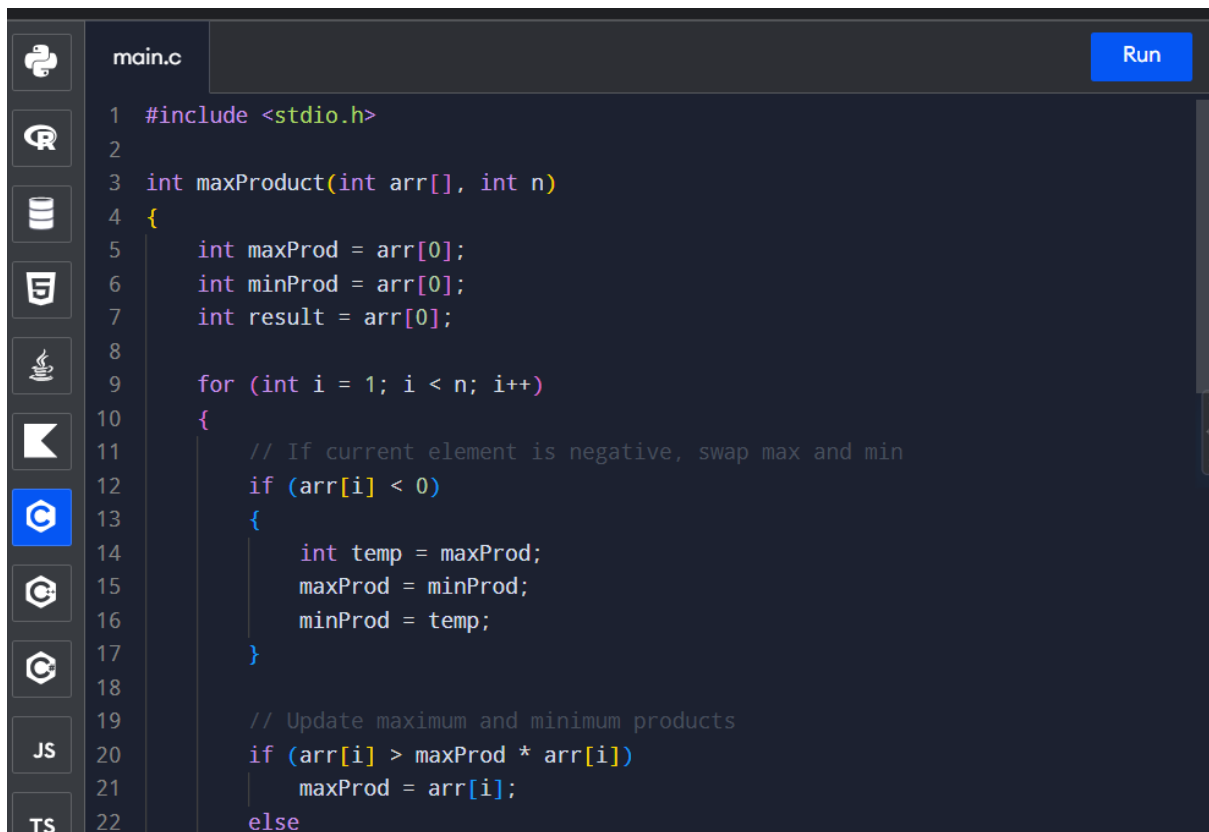
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CODE CHALLENGE

Q33 – Maximum Product Subarray using Dynamic Programming:

CODE



The image shows a code editor window with a dark theme. The title bar at the top says "main.c" and there is a blue "Run" button on the right. On the left side, there is a vertical toolbar with icons for Python, R, SQL, Java, C++, JavaScript, and TypeScript. The C++ icon is currently selected and highlighted in blue. The main area of the editor contains C++ code for finding the maximum product subarray. The code is as follows:

```
1  #include <stdio.h>
2
3  int maxProduct(int arr[], int n)
4  {
5      int maxProd = arr[0];
6      int minProd = arr[0];
7      int result = arr[0];
8
9      for (int i = 1; i < n; i++)
10     {
11         // If current element is negative, swap max and min
12         if (arr[i] < 0)
13         {
14             int temp = maxProd;
15             maxProd = minProd;
16             minProd = temp;
17         }
18
19         // Update maximum and minimum products
20         if (arr[i] > maxProd * arr[i])
21             maxProd = arr[i];
22         else
```

```
main.c Run
4 {
10 {
22     else
23         maxProd = maxProd * arr[i];
24
25         if (arr[i] < minProd * arr[i])
26             minProd = arr[i];
27         else
28             minProd = minProd * arr[i];
29
30         // Update overall maximum
31         if (maxProd > result)
32             result = maxProd;
33     }
34
35     return result;
36 }
37
38 int main()
39 {
40     int n;
41
```

```
main.c Run
37
38 int main()
39 {
40     int n;
41
42     printf("Enter number of elements: ");
43     scanf("%d", &n);
44
45     int arr[n];
46
47     printf("Enter array elements: ");
48     for (int i = 0; i < n; i++)
49     {
50         scanf("%d", &arr[i]);
51     }
52
53     int result = maxProduct(arr, n);
54
55     printf("Max Product = %d\n", result);
56
57     return 0;
58 }
```

OUTPUT

Output



```
Enter number of elements: 5  
Enter array elements: 2 3 -2 4 -1  
Max Product = 48
```

```
=== Code Execution Successful ===
```